AIGC 5500 | ADVANCED DEEP LEARNING | MIDTERM PROJECT

Does built-in structure help spam detection?

A CNN, an LSTM, and a Transformer against a bag-of-embeddings baseline

Baseline | CNN | LSTM | Transformer

Team MVPs

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Professor: Hossein Pourmodheji | Humber Polytechnic | November 2026

https://github.com/erictsaily/AIGC5500-Midterm-Project

THE QUESTION

Which built-in structure helps classify spam?

Baseline

No structure

Averages word embeddings: knows which words appear

CNN

Local patterns

Sliding windows of 3, 5, and 7 words

LSTM

Sequential state

Reads word by word, keeps one state

Transformer

Direct lookup

Any word attends to any other word

Does any structure beat a model with none, and why?

Evidence: four models with three seeds each | one ablation per family | a second dataset | inference cost

Enron spam, audited before splitting

Source
31,716 train + 2,000 test emails

↓

Removed before splitting
51 empty | 3,137 duplicates
372 train-test overlaps | 19 test duplicates

↓

Split
Stratified 80/20 of development data, seed 42

Partition	Emails	Ham	Spam
Train	22,524	11,681 (51.9%)	10,843 (48.1%)
Validation	5,632	2,921 (51.9%)	2,711 (48.1%)
Test	1,981	*Held out, scored once at the end*	

Every decision uses validation
Vocabulary, checkpoints, and model choice never see the test split

Source: notebooks/01_data_preparation.ipynb, sections 2.1 to 2.6 (dataset: SetFit/enron_spam on Hugging Face)

IDENTICAL CONDITIONS

Same inputs, budget, and selection rule for every model

Input

- Lowercased words and punctuation
- 10,000-token vocabulary from training text only
- 256-token limit: **32.2%** of emails cut, **49.6%** of tokens dropped

Training

- AdamW, learning rate 0.001, weight decay 0.01
- Gradient clip 1.0, batch 64, **10 epochs**
- Best epoch by **validation spam F1**
- Seeds 42, 123, 2026 for every run
- No hyperparameter search for any model

Model size

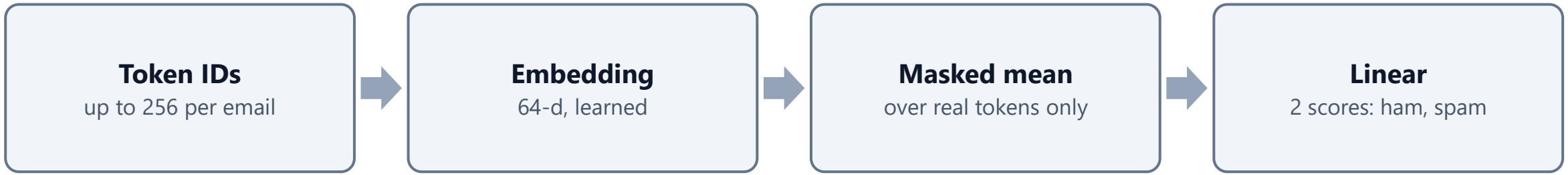
Model	Parameters
Baseline	640,130
CNN	736,902
LSTM	739,586
Transformer	723,586

Largest / smallest = 1.16

Every model has the same 640,000-weight embedding table

Source: src/config.py, src/text.py, and the parameter counts printed in notebooks 02 to 05

Baseline: an average of word embeddings



- Equivalent to a learned **spam score per word**, averaged
- Knows **which words appear**, not their order
- Same data, budget, seeds, and tuning (none) as the deep models

640,130

parameters

trained from scratch, no pretrained weights

Source: src/baseline_model.py, notebooks/02_baseline_experiment.ipynb

CNN: phrase detectors with max-pooling



- Shared weights: a filter finds its phrase **anywhere**
- Max-pool asks: **did this phrase appear at all?**
- Three widths catch phrases of different lengths

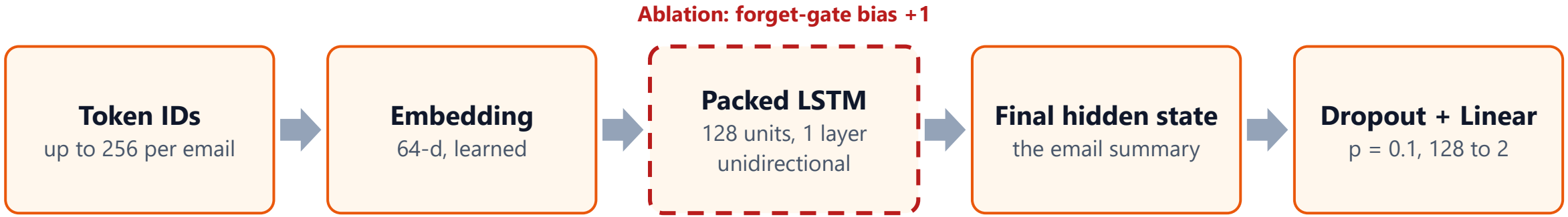
736,902

parameters

trained from scratch, no pretrained weights

Source: src/cnn_model.py, notebooks/03_cnn_experiment.ipynb

LSTM: one sequential state, read word by word



- Gates decide what the cell state **keeps or forgets**
- Padding never enters the state (verified in the notebook)
- The classifier sees **only the final state**

739,586

parameters

trained from scratch, no pretrained weights

Source: src/lstm_model.py, notebooks/04_lstm_experiment.ipynb

Transformer: every word can look up every other



- **Direct lookup** in one step: no state, no window
- Positions are added because attention alone is **order-blind**
- Padding mask in attention and in pooling

723,586

parameters

trained from scratch, no pretrained weights

Source: src/transformer_model.py, notebooks/05_transformer_experiment.ipynb

Baseline leads on both splits; the LSTM trails

Model	Validation spam F1 (selection)				Test spam F1 (scored once)			
	seed 42	seed 123	seed 2026	mean ± SD	seed 42	seed 123	seed 2026	mean ± SD
Baseline	**0.9888**	**0.9882**	**0.9890**	**0.9887 ± 0.0004**	**0.9890**	**0.9910**	**0.9900**	**0.9900 ± 0.0010**
CNN	0.9874	0.9868	0.9871	0.9871 ± 0.0003	0.9889	0.9885	0.9889	0.9888 ± 0.0002
LSTM	0.9834	0.9853	0.9810	0.9832 ± 0.0022	0.9854	0.9820	0.9780	0.9818 ± 0.0037
Transformer	0.9882	0.9851	0.9854	0.9862 ± 0.0017	0.9849	0.9859	0.9868	0.9859 ± 0.0010

Mean test errors (of 1,981 emails): Baseline 20 | CNN 22 | Transformer 28 | LSTM 36

Spam F1 (spam = positive class). Mean ± sample SD over three seeds; best in each column in bold. Source: results/enron//validation_per_seed.csv and test_per_seed.csv*

EVIDENCE

Every number comes from notebook output

03_cnn_experiment.ipynb, section 4 (printed output)

```
CNN (full) seed 42 best epoch 6 val acc 0.9879 F1 0.9874 (82s)
CNN (full) seed 123 best epoch 6 val acc 0.9872 F1 0.9868 (82s)
CNN (full) seed 2026 best epoch 10 val acc 0.9876 F1 0.9871 (83s)
Ablation: kernel width 1 seed 42 best epoch 7 val acc 0.9893 F1 0.9890 (52s)
Ablation: kernel width 1 seed 123 best epoch 9 val acc 0.9897 F1 0.9894 (52s)
Ablation: kernel width 1 seed 2026 best epoch 10 val acc 0.9895 F1 0.9891 (52s)
```

05_transformer_experiment.ipynb, section 4 (printed output)

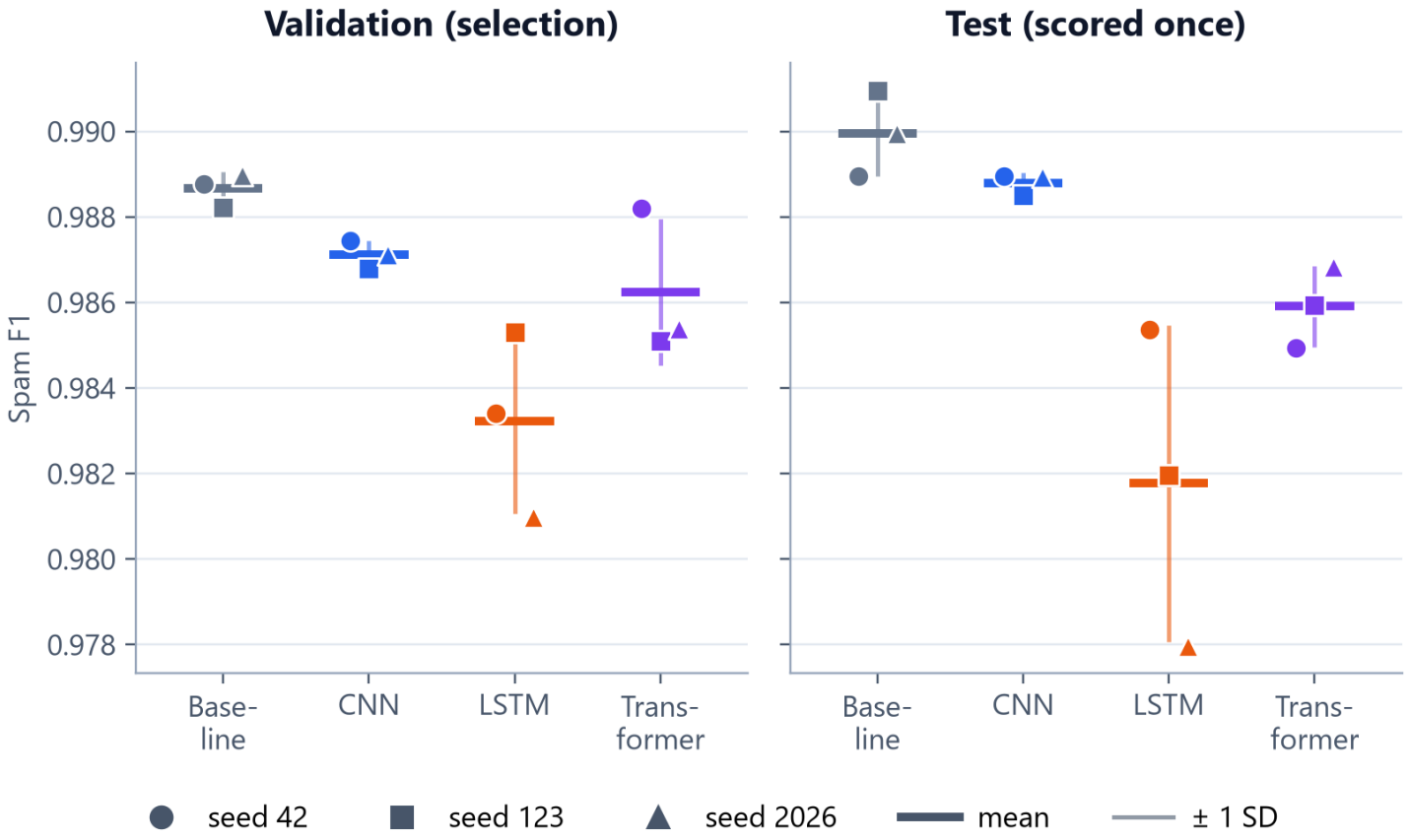
```
Transformer (full) seed 42 best epoch 9 val acc 0.9886 F1 0.9882 (82s)
Transformer (full) seed 123 best epoch 8 val acc 0.9856 F1 0.9851 (82s)
Transformer (full) seed 2026 best epoch 10 val acc 0.9860 F1 0.9854 (82s)
Ablation: no positions seed 42 best epoch 8 val acc 0.9879 F1 0.9875 (82s)
Ablation: no positions seed 123 best epoch 10 val acc 0.9874 F1 0.9870 (82s)
Ablation: no positions seed 2026 best epoch 10 val acc 0.9863 F1 0.9859 (81s)
```

LIVE: switch to the notebook (about 30 s)

Where each number is written

- Baseline: notebook 02, sections 5 and 7
- CNN: notebook 03, sections 4 and 7
- LSTM: notebook 04, sections 5 and 6
- Transformer: notebook 05, sections 4 and 7
- Second dataset: notebook 06
- Tables: built from the CSV files in results/

Pooling models tie; the final-state LSTM is the outlier



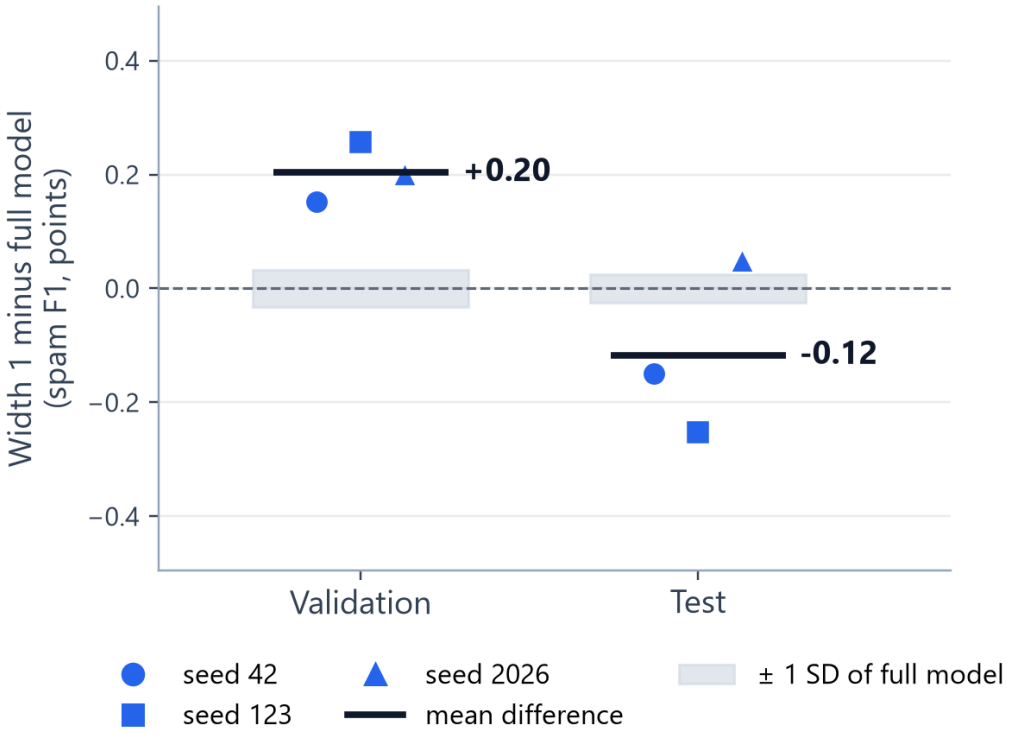
- Baseline, CNN, and Transformer pool evidence from **every position**: within 0.0024 on validation
- Spam is largely **lexical**: which words appear
- LSTM: a cue must survive **up to 256 updates** to reach the classifier, giving the lowest mean and widest spread

An explanation consistent with the evidence, not tested directly.

Source: same per-seed CSV files as the results table

Width-1 kernels: higher on validation, within noise on test

Seed	Validation		Test	
	full	width 1	full	width 1
42	0.9874	0.9890	0.9889	0.9874
123	0.9868	0.9894	0.9885	0.9860
2026	0.9871	0.9891	0.9889	0.9894
Mean ± SD	**0.9871 ± 0.0003**	**0.9892 ± 0.0002**	**0.9888 ± 0.0002**	**0.9876 ± 0.0017**
Difference	**+0.0020**		**-0.0012**	

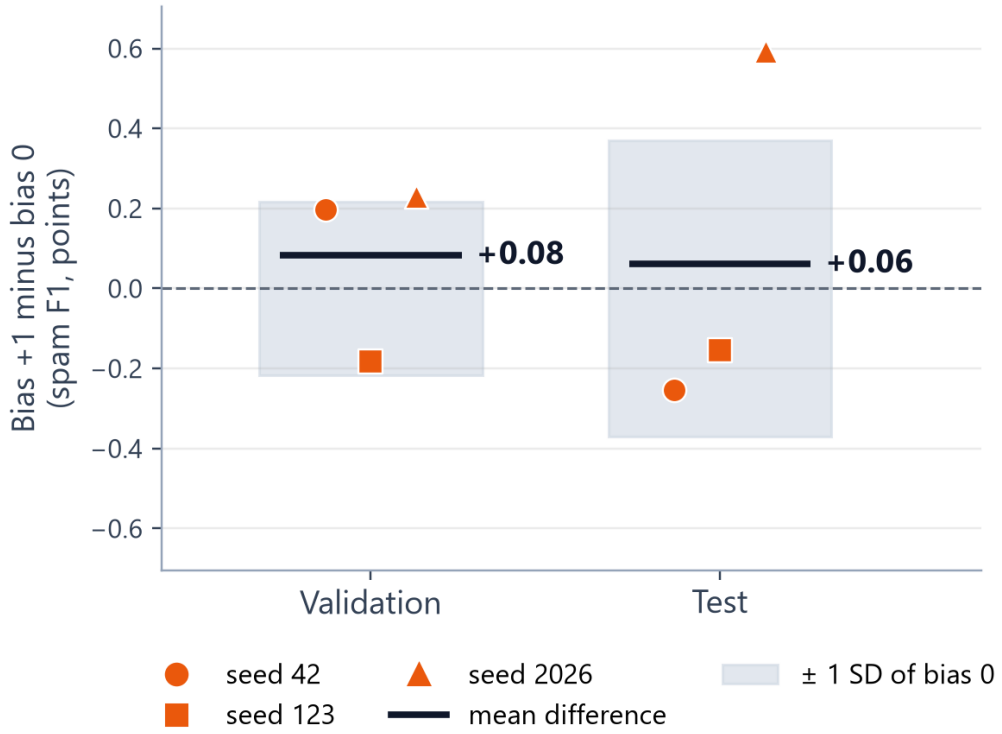


Multi-word windows did not help. Width 1: 660,102 parameters, about 37% less training time per run.

Spam F1; mean ± sample SD over seeds 42, 123, 2026. Shaded band: ± 1 SD of the full model across seeds. Source: results/enron/cnn/

Forget-gate bias +1: no reliable effect

Seed	Validation		Test	
	bias 0	bias +1	bias 0	bias +1
42	0.9834	0.9854	0.9854	0.9828
123	0.9853	0.9835	0.9820	0.9804
2026	0.9810	0.9833	0.9780	0.9839
Mean ± SD	**0.9832 ± 0.0022**	**0.9841 ± 0.0012**	**0.9818 ± 0.0037**	**0.9824 ± 0.0018**
Difference	**+0.0008**		**+0.0006**	

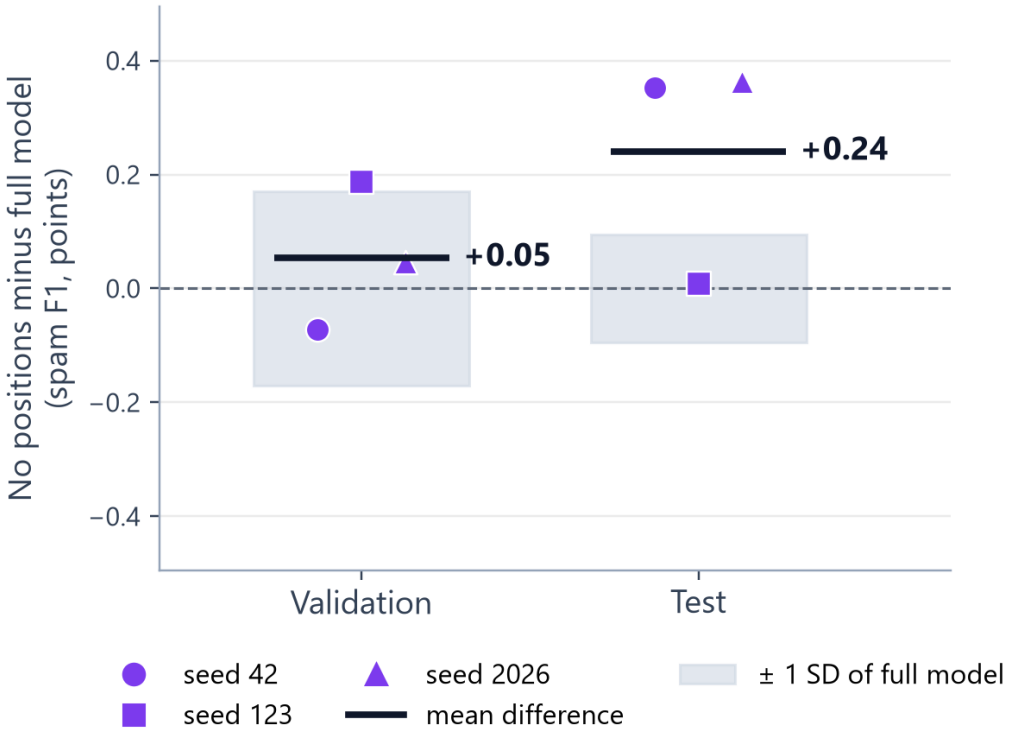


Mixed seeds on both splits. Spread across seeds halves, a hint only. Same 739,586 parameters.

Spam F1; mean ± sample SD over seeds 42, 123, 2026. Shaded band: ± 1 SD of the bias 0 model across seeds. Source: results/enron/lstm/

Removing positions: no measurable cost

Seed	Validation		Test	
	full	no positions	full	no positions
42	0.9882	0.9875	0.9849	0.9885
123	0.9851	0.9870	0.9859	0.9860
2026	0.9854	0.9859	0.9868	0.9905
Mean ± SD	**0.9862 ± 0.0017**	**0.9868 ± 0.0008**	**0.9859 ± 0.0010**	**0.9883 ± 0.0022**
Difference	**+0.0005**		**+0.0024**	



Order information gave no benefit; the test gain of about 5 emails is close to the seed spread.

Spam F1; mean ± sample SD over seeds 42, 123, 2026. Shaded band: ± 1 SD of the full model across seeds. Source: results/enron/transformer/

ABLATIONS TOGETHER

Removing word order did not measurably hurt

CNN

Width-1 kernels

Order-free windows

+0.0020 validation

Transformer

No positions

Order-free set of words

+0.0005 validation

Baseline

Averaged embeddings

Order-free by design

best on validation and test

The useful signal is which words appear, not how they are arranged

The LSTM ablation tests optimisation at initialisation, not order: its null result points to the final-state architecture.

Kaggle ham/spam: a harder dataset, the same protocol

20,000 emails, 47% spam

Split 14,381 / 3,596 / 2,000

Own 10,000-token vocab

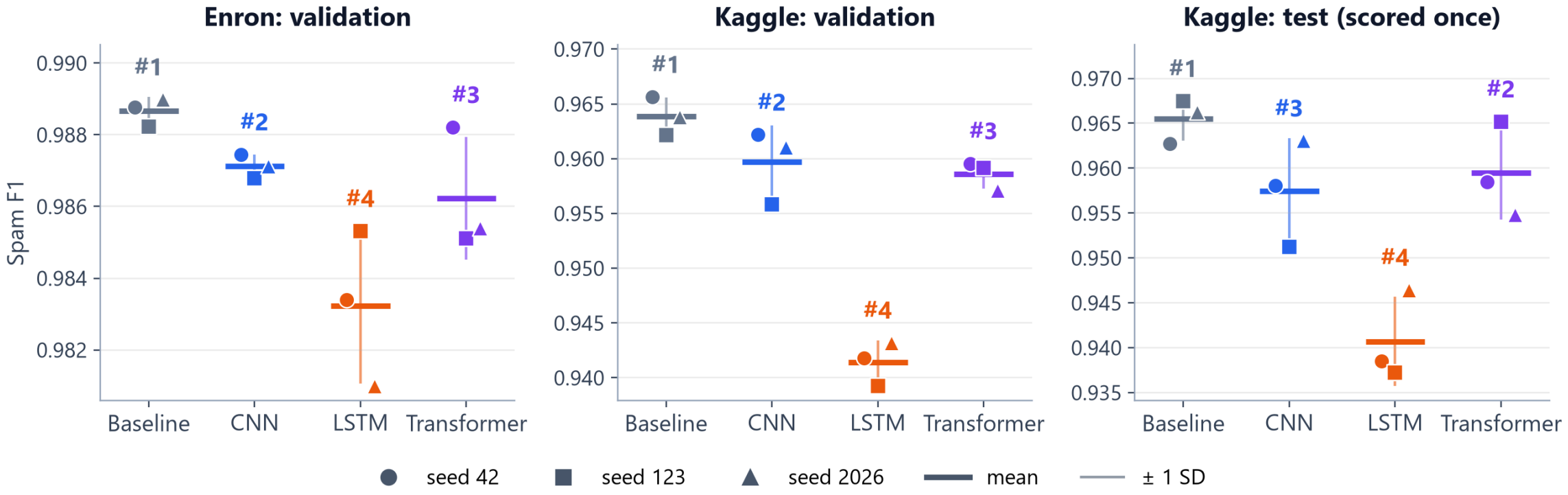
Same protocol, same seeds

Model	Validation spam F1 (selection)				Test spam F1 (scored once)			
	seed 42	seed 123	seed 2026	mean ± SD	seed 42	seed 123	seed 2026	mean ± SD
Baseline	**0.9656**	**0.9622**	**0.9638**	**0.9639 ± 0.0017**	**0.9628**	**0.9675**	**0.9662**	**0.9655 ± 0.0024**
CNN	0.9622	0.9558	0.9610	0.9597 ± 0.0034	0.9580	0.9512	0.9630	0.9574 ± 0.0059
LSTM	0.9418	0.9392	0.9432	0.9414 ± 0.0020	0.9385	0.9372	0.9464	0.9407 ± 0.0050
Transformer	0.9595	0.9591	0.9571	0.9586 ± 0.0013	0.9585	0.9652	0.9548	0.9595 ± 0.0052

Spam F1; mean ± sample SD over three seeds; best in each column in bold. Source: results/kaggle// and notebooks/06_second_dataset_experiment.ipynb*

DOES IT HOLD?

Same validation ranking on both datasets; the LSTM gap widens



Ranking
Baseline, CNN, Transformer, LSTM on both validation sets

LSTM gap to baseline
0.54 points on Enron, 2.25 on Kaggle

Kaggle test
CNN and Transformer swap by 0.0021, inside the seed spread

Labels show the rank of each mean. Source: results/enron/ and results/kaggle/

INFERENCE COST

Similar sizes; timings are not comparable across GPUs

Model	Parameters	ms per email (batch 64)	Peak GPU memory	Hardware
Baseline	640,130	0.036	23.7 MB peak	RTX 3060 Laptop GPU
CNN	736,902	0.098	72.5 MB peak	GTX 1050 Ti
LSTM	739,586	0.120	43.2 MiB extra	RTX 3060 Laptop GPU
Transformer	723,586	0.144	181.1 MB peak	Tesla T4

Not a ranking

Three GPUs and two procedures; the baseline and LSTM share a GPU, not a procedure

Valid within a family, same GPU

CNN width 1: 41% less time, 38% less memory. Transformer without positions: about the same.

Time per email = batch-of-64 time / 64. Peak: CUDA peak allocated incl. weights. Extra: peak above weights and inputs. Source: results/enron//inference_cost.csv*

LIMITATIONS

What we would not claim

"The CNN is best"	The baseline beats it on validation and test; the top three are a few test emails apart
"Order never matters"	Two email datasets only, and the 256-token cut drops about half the tokens
"LSTMs are worse"	One configuration; two of three runs peaked at the final epoch, 10
"Small gaps are real"	Three seeds; gaps below about 0.002 are within noise
"Each model at its best"	Equal but minimal tuning; Enron is near the ceiling (baseline test 0.990)

Who owned what, and what we did not write

Member	Owned piece	Ablation
Daniel Castillo	Data preparation, LSTM, baseline test	Forget-gate bias
Nouroz Ali	Baseline	-
Bryan Jacome	CNN	Kernel width 1
Abhi Patel	Transformer	No positions
Eric Ly	Second dataset	-

References

- Metsis, Androutsopoulos, & Paliouras (2006). Spam filtering with naive Bayes: which naive Bayes? CEAS.
- SetFit (n.d.). enron_spam [Data set]. Hugging Face.
- artofintelligence (n.d.). ham-spam [Data set]. Kaggle.
- Kim (2014). Convolutional neural networks for sentence classification. EMNLP.
- Hochreiter & Schmidhuber (1997). Long short-term memory. Neural Computation.
- Jozefowicz, Zaremba, & Sutskever (2015). An empirical exploration of recurrent network architectures. ICML.
- Vaswani et al. (2017). Attention is all you need. NeurIPS.
- Xiong et al. (2020). On layer normalization in the Transformer architecture. ICML.
- Loshchilov & Hutter (2019). Decoupled weight decay regularization. ICLR.
- Paszke et al. (2019). PyTorch. NeurIPS; PyTorch documentation.
- AI assistance: ChatGPT (notebook starter structure); Cursor AI agent (draft slides and script, reviewed by the team).

ANTICIPATED QUESTIONS

Three questions a sceptic should ask

Is the baseline winning only because the deep models were under-trained or under-tuned?

The CNN peaked at epoch 6 on two seeds; the baseline at 9 or 10. Only the LSTM peaked at the end. Claim: at equal, minimal tuning.

Did the 256-token cut remove the signal that order-aware models need?

Cannot rule it out: about two thirds of emails fit whole, but long emails were never scored separately. Next test: a longer cap.

With gaps this small, which parts of the ranking are real?

Every LSTM seed is below every baseline seed on both datasets; width 1 beats the full CNN on every seed. The top-three order is not: the CNN and Transformer swap on the Kaggle test.

Repository: https://github.com/erictsaily/AIGC5500-Midterm-Project